



LEgoarch

Name a building. Get a buildable LEGO set.

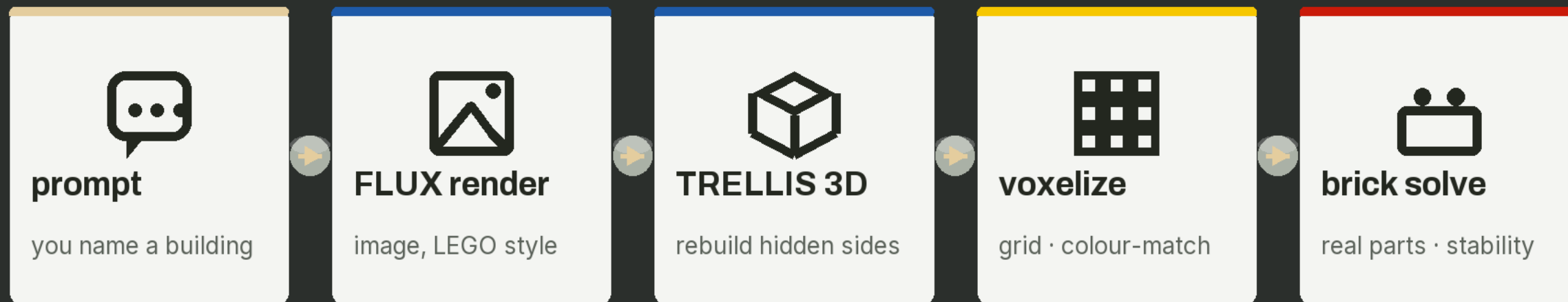
Generative AI proposes the form · deterministic computation proves it's buildable.



HOW IT WORKS

Three steps. One build.

A model proposes the shape; plain code proves you can actually build it.



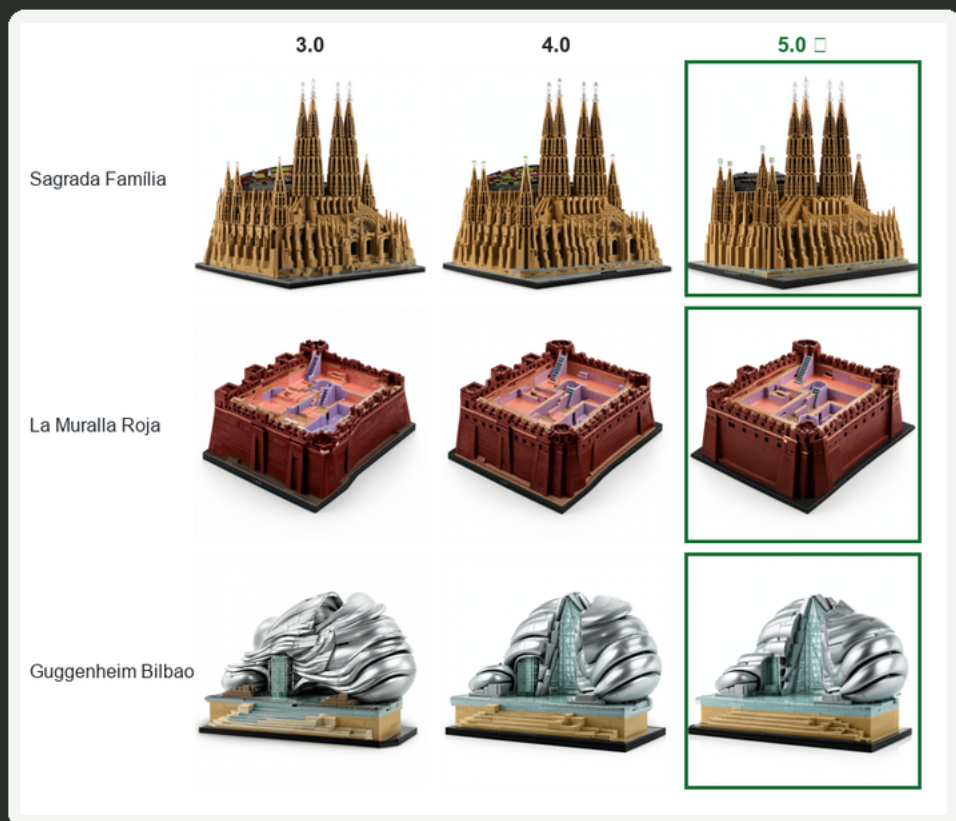
blue = generative · yellow = deterministic compute · red = the real catalog





WE TUNED THIS BY HAND

Getting the render right.



same three buildings, one setting swept at a time — winner ringed



STEPS

28

GUIDANCE

5.0

LORA

1.0

NEGATIVE

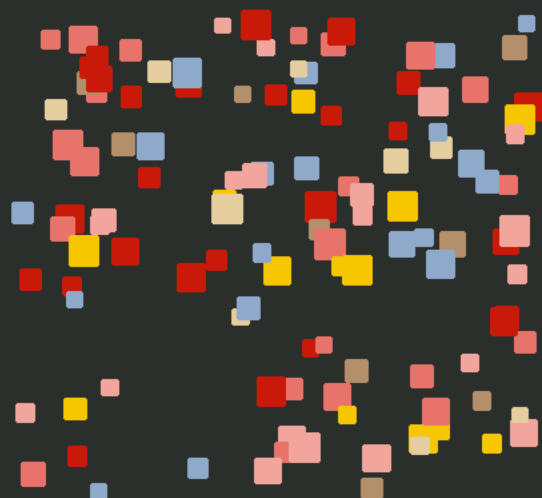
on

Turn the LoRA down and the studs literally fade away — that's where the LEGO look lives. The prompt itself is an 8-part grammar that keeps the shape fused so it survives the 3D step.





Fewer bricks, same building.



every speck = its own 1x1
brick



merged onto the true colour

up to

-46%

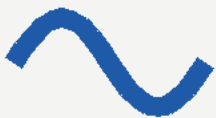
fewer pieces · -19% on La Muralla

WHY FEWER IS GOOD

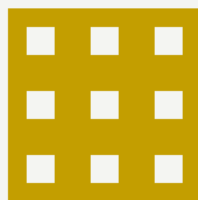
cheaper · simpler to build · no
loss of stability



And here's where it falls apart.



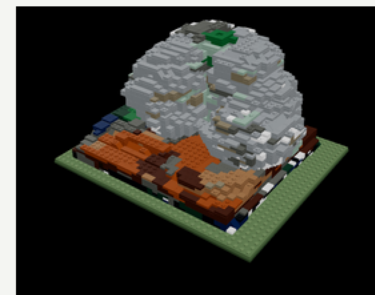
smooth Gehry curves



forced onto a grid



a blob — not Bilbao



Still structurally sound — connected, 93% supported. It just isn't legible.

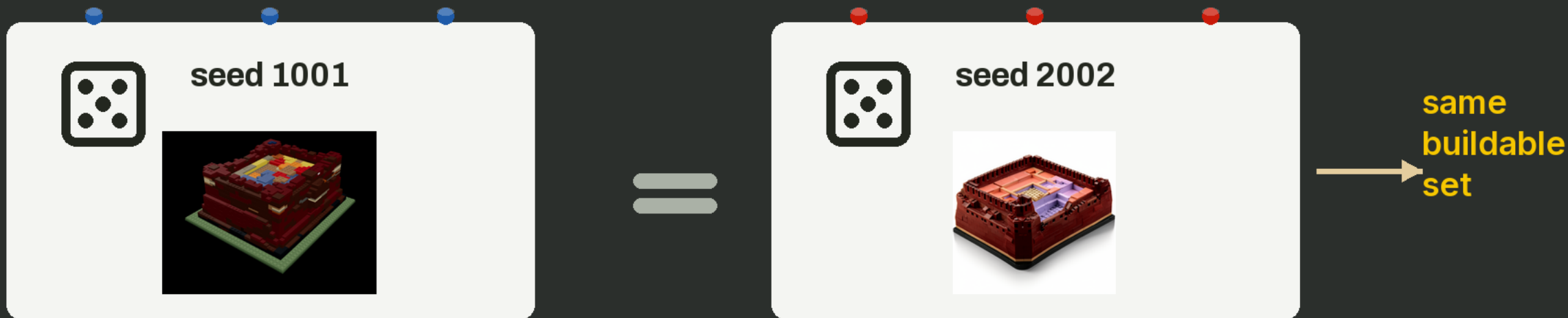
Voxels can't hold a curve. We like that we know exactly where the method ends.





THIS SURPRISED US

Same prompt, same set.



A new seed is a different generation — the pieces vary — but it's always a single, connected, recognizable set. The result is the method, not a lucky seed.





BUILDABLE MEANS ORDERABLE

Real parts. Three sizes.



48 LEGO colours

REAL PARTS

44

VALIDATED COMBOS

1,598

one model · three sizes



24

32

48

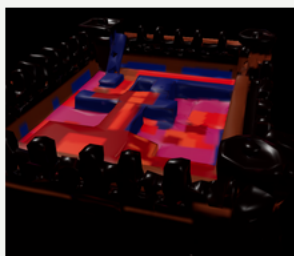
≈ 2k draft · 4.7k set · 22k flagship



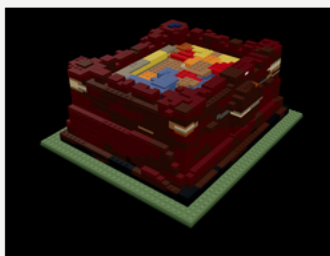


FROM A SENTENCE TO A SET

Everything you get.



3D mesh



legolized build

add screenshot



the boxed set

add screenshot



instruction booklet

add screenshot



parts list + price

add screenshot



your shelf





IEgoarCh

From a sentence to a set you can build.

Emilie Chidiac · Charles · Generative AI · 2026